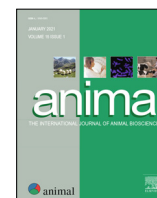




Animal

The international journal of animal biosciences



Modulation of intestinal morphology and microbiota by dietary *Macleaya cordata* extract supplementation in Xuefeng Black-boned Chicken



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ARTICLE INFO

Article history:

Received 21 February 2021

Revised 29 September 2021

Accepted 4 October 2021

Available online 10 November 2021

Keywords:

Gut morphological improvement

Herbal extract antibiotics

Indigenous chicken

Intestinal microbial diversity

Microbiota composition

ABSTRACT

Antibiotics are commonly overused to prevent livestock from diseases and to increase production performance. As potential substitutes of antibiotics, plant extracts have attracted the attention of researchers. It was known to all that addition of *Macleaya cordata* extract (MCE) to the food could advance immunity, intestinal health and animal performance. Thus, it was conducted to investigate the influence of MCE (0, 100, 150 and 200 mg/kg, with six replicate pens/treatment and 24 hens/pen) on intestinal morphology and microbial diversity in different intestinal segments in Xuefeng black-boned chicken in this study. The results showed that MCE supplement (100, 150 and 200 mg/kg) significantly diminished ($P < 0.05$) the crypt depth of the jejunum as compared to basal diet group. The 100 mg/kg group displayed a marked increase ($P < 0.05$), compared with 0 and 200 mg/kg group, in ileum microbial diversity as represented by the Shannon's index. In the cecum, treatment of MCE significantly decreased ($P < 0.01$) the Firmicutes, but Deferribacteres in 200 mg/kg MCE group were significantly raised ($P < 0.05$). In conclusion, we found that MCE improved intestinal morphology and reduced the crypt depth in jejunum. Together, addition of 200 mg/kg MCE modulated intestinal microbiota, increased beneficial bacteria such as *Lactobacillus*. Adding 100 mg/kg MCE to diet increased bacterial community diversity and relative abundance in jejunum and ileum, but had no effect on cecum microbial diversity.

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Implications

This study indicated that *Macleaya cordata* extract could improve intestinal morphology and gut microbiota composition of Xuefeng black-boned chicken evidenced by increased beneficial bacteria, increased bacterial community diversity and relative abundance in jejunum and ileum. *Macleaya cordata* extract could be regarded as antibiotic substitutes. These findings would provide new insights into suitable alternatives to antibiotics in poultry production and provide an evidence basis upon which to advance research on *Macleaya cordata* extract.

Introduction

Observational data achieved during the previous decades advised that the intestinal of poultry was an open ecosystem, which was occupied by a great deal of microscopic organisms and formed a microecological environment with mixed functions (Fan and Pedersen, 2021). Intestinal microorganisms provided a key link to the host, such as digesting food, regulating gut endocrine function, improving drug action and metabolism, discarding toxins (Lynch and Pedersen, 2016). Imbalance of intestinal flora or decrease of microbial diversity will lead to the decrease of immunity and nutrient utilization, which might be related to the occurrence of chylous diarrhoea and inflammatory bowel disease (Kamada et al., 2013). Thus, a healthy intestinal flora was conducive to maintaining normal physiological function of the host and giving full play to its growth or production potential.

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Macleaya cordata, being rich in alkaloids, was a perennial herb. Sanguinarine and chelerythrine, the key compositions of isoquinoline alkaloids (IQAs) originated from *Macleaya cordata* extract (MCE), have been reported to involve in regulating anti-inflammation (Chaturvedi et al., 1997; Wullaert et al., 2011; Niu et al., 2012; Wang et al., 2018; Xue et al., 2017), modulating intestinal microbial (Hamoud et al., 2015), mediating immune response (Bavarsadi et al., 2016), and promoting growth in livestock farming. The recent literature has demonstrated that supplement of MCE caused positive effects on improving intestinal health, for example, *Macleaya cordata*-derived alkaloid extract reduced damage of the duodenum, jejunum, and ileum (Xue et al., 2017).

Indigenous chicken breeds meets growing and changing demand for food diversification in China, and its rate of market share showed a steady increase. Xuefeng black-bone chicken, one of the indigenous chicken breeds, has been presented in the record of National Livestock and Poultry Genetic Resources Protection of China. As the meat- and egg-type regional chicken breed, it owned considerable reputation due to its nutritional value and medicinal properties (Liu et al., 2016b). In our previous studies, no significant differences were observed in feed intake and egg production (Liu et al., 2020), but MCE improved egg quality, laying performance, immune activity and anti-oxidation indices remarkably in Xuefeng black-bone chicken (Guo et al., 2021). Given the results outlined above, it was speculated that MCE might affect multiple performance of Xuefeng black-bone chicken by mediating intestinal flora and microbial diversity. However, how MCE affect intestinal microbe of native chicken of China still remained to be discovered. This study was operated to explore the influences of MCE on intestinal morphology and microbial diversity in different intestinal segments of the parental population, and to provide a basis for MCE application in production of Xuefeng black-boned chicken.

Material and methods

Animals and experimental design

Five hundred and seventy-six birds (47 weeks old, white feathers) were randomly assigned to four groups with six replicas (24 hens/replica). MCE was added to diet at the level of 0 mg/kg (group A), 100 mg/kg (group B), 150 mg/kg (group C) and 200 mg/kg (group D), respectively, and fed for 84 d. The feeding experiment was the same as our previous study (Guo et al., 2021).

Sample collection

At the end of the test, chickens were randomly chosen and sacrificed by cervical dislocation. After slaughter, the digesta samples were from jejunum, ileum and cecum, they were collected in sterilized plastic tubes and stored immediately in liquid nitrogen to investigate intestinal microflora. Additionally, segments of the middle of jejunum and ileum were excised and fixed in 4% paraformaldehyde for more than 24 h to make intestinal sections.

Intestinal morphology

All intestinal segments from each jejunum and ileum were embedded after removing the fixation solution, using standard paraffin embedding procedures. Subsequently, a 5- μ m section of each sample was located on a glass slide and stained by haematoxylin and eosin (H&E) for light microscopy. Microscopic examinations were conducted using a Zeiss microscope (Axio Imager, A1), and photographs were taken with MShot Image Analysis System. Five typical fields were used to measure the crypt depth and

the villus height for estimating the average of the three values and the proportion of villi height to crypt depth.

Intestinal microflora

Bacterial DNA extraction and detection

The digesta samples of three different segments (jejunum, ileum and cecum) from four groups were used to extract genomic DNA by using KIT (DP328, Tiangen Biochemical Technology Co., Ltd., China). The qualified DNA samples were dispatched to Shanghai Majorbio Bio-pharm Technology Co., Ltd (China) for PCR amplification and sequencing analysis.

Illumina MiSeq high-throughput sequencing

Using quantified DNA as template, the V3-V4 regions of 16S rRNA gene were magnified with primer pairs 338F (5'-ACTCC TACGGGAGGCAGCAG-3') and 806R (5'-GGACTACHVGGGTWTC TAAT-3'). Purified amplicons were combined in equimolar and paired-end sequenced on a NovaSeq PE250 platform (Illumina, San Diego, USA) in response to the standard protocols by Majorbio Bio-Pharm Technology Co. Ltd. (Shanghai, China).

Sequence processing (Dong et al., 2021)

The original sequences were demultiplexed, quality-filtered using fastp version 0.20.0 (Chen et al., 2018a; 2018b) and merged using FLASH version 1.2.7 (Magoc and Salzberg, 2011). The criteria were as following: a. the 300 bp reads were detruncated at any site obtaining an average quality score of <20 over a 50 bp sliding window, and the detruncated reads shorter than 50 bp were discarded, reads containing equivocal characters were also discarded; b. only overlapping sequences greater than 10 bp were assembled according to their overlapped sequence. The maximum mismatch ratio in the overlap region is 0.2. Reads that could not be assembled were scraped; c. samples were detected according to the barcode and primers, and the sequence direction was altered. The barcode matching was accurate, and the two nucleotides in the primer matching did not match. Operational taxonomic units (OTUs) with a similarity cut-off rate of 97% (Edgar, 2013) were clustered using UPARSE version 7.1. RDP Classifier Version 2.2 was used to classify each OTU representative sequence, and Silva V132 was used for classification and identification.

Statistical analysis

Results were expressed as mean and SEM. Experimental data were analyzed with one-way ANOVA with SPSS 22.0 software (SPSS Inc., Chicago, IL). The intestinal microbial data were analyzed using Kruskal-Wallis's test in Majorbio Cloud Platform (<http://www.majorbio.com>). Significance was expressed as * $P < 0.05$ and ** $P < 0.01$.

Results

Intestinal morphology

The intestinal morphology was observed by measuring the crypt depth (CD), villus height (VH), and the proportion of villi height to crypt depth (VH/CD) (Table 1). The histopathological analysis of intestinal mucosal sections of the jejunum demonstrated that CD of the groups with the MCE supplementation (groups B, C and D) were significantly lesser than that of the control ($P < 0.05$). Among four groups, VH/CD of 200 mg/kg group was the highest ($P > 0.05$) in both of jejunum and ileum.

Table 1
Influences of MCE on intestinal morphology of Xuefeng black-boned chicken.¹

MCE mg/kg	0	100	150	200	SEM	P-value
Jejunum						
VH (μm)	856.07	690.53	722.73	754.08	29.41	0.223
CD (μm)	222.76 ^a	172.70 ^b	177.02 ^b	174.28 ^b	7.78	0.030
VH/CD	3.86	3.99	4.08	4.34	0.11	0.544
Ileum						
VH (μm)	748.90	650.16	658.58	706.42	26.53	0.587
CD (μm)	219.28	170.17	177.34	164.28	11.57	0.360
VH/CD	3.46	3.87	3.78	4.37	0.15	0.155

Abbreviations: VH: Villus height; CD: Crypt depth; MCE: *Macleaya cordata* extract.

^{a,b} Means in the same row with no common superscript differ significantly ($P < 0.05$).

¹ Values were represented as mean of three birds.

Intestinal microbial diversity

A total of 36 intestinal content samples were collected for DNA extraction and 16S rDNA sequencing. The sequencing data were filtered to gain valid data of 1 943 348 tags; the average length was about 423.6 bp tags (Fig. 1).

Operational taxonomic unit analysis

The valid tags of all samples were clustered and the sequences with more than 97% similarity were considered as an OTU, and the similar OTUs were intersected in the Venn diagram. The Venn diagram (Fig. 2) showed the specific OTUs in different intestinal sections of different groups (a: jejunum, b: ileum; c: cecum). In groups A, B, C and D, the number of OTUs of the jejunum was 689, 812, 760 and 330, respectively (Fig. 2a). However, only 176 OTUs were shared across the four groups. Among the pairwise analysis, only 257 OTUs were shared in the groups A and D, being the smallest number of shared OTUs. As to the number of the species unique to each group, group B presented the largest unique OTUs (259) than others.

In ileum, the number of OTUs detected in groups A, B, C and D was 529, 853, 549 and 347, respectively (Fig. 2b). Furthermore, analysis indicated that the number of OTUs shared by four groups was 159, and the number of OTUs shared by groups B, C and D is 192. However, only 218 OTUs were shared between group A and group D, being the least number in all pairwise groups. With the number of 340, the group B showed the highest number of unique OTUs, compared with the other three groups of MCE supplement. As shown in Fig. 2c, the number of OTUs detected in cecum of groups A, B, C and D was 866, 754 834 and 780, respectively. A total

of 570 common OTUs were identified across four groups in cecum, and the OTUs shared by groups B, C and D were 599. Although 660 OTUs were shared by group A and group B, being the smallest one among all pairwise comparison, the largest number of the unique OTUs was observed in group A.

Shannon index

We performed an analysis of the differences between groups for different intestinal segments using Shannon index (Fig. 2). No significant difference in Shannon index was found in jejunum ($P > 0.05$), in treatment groups compared with group A (Fig. 2d). Nevertheless, we observed a higher Shannon index in group B in comparison to group A ($P > 0.05$). As shown in Fig. 2e, in ileum, Shannon index of group B was significantly higher than groups A and D ($P < 0.05$). These results suggested that daily addition of 100 mg/kg MCE might have beneficial effect on microbiome diversity in ileum.

Rarefaction curve

The data collected were analysed, and rarefaction curves were plotted (Fig. 3). The graph line reached an asymptote with x axis which indicated that the detection ratio of the sample microbial community was close to saturation, and almost all the species in this area were sampled.

Heat map analysis-phyllum level

Same species was clustered in Heat map, and the microbial population was remarked by the shade of colour. At the phylum level,

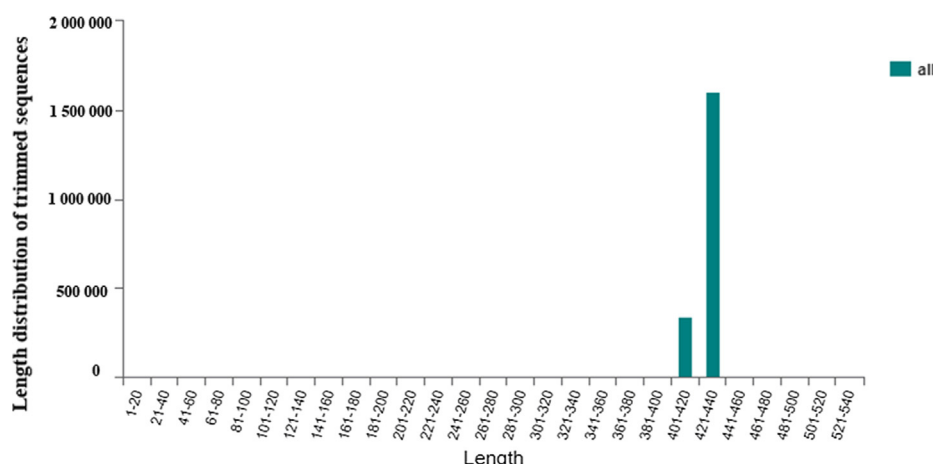


Fig. 1. The length distribution of the optimised sequence in intestinal content samples when Xuefeng black-boned chicken were fed different *Macleaya cordata* extract.

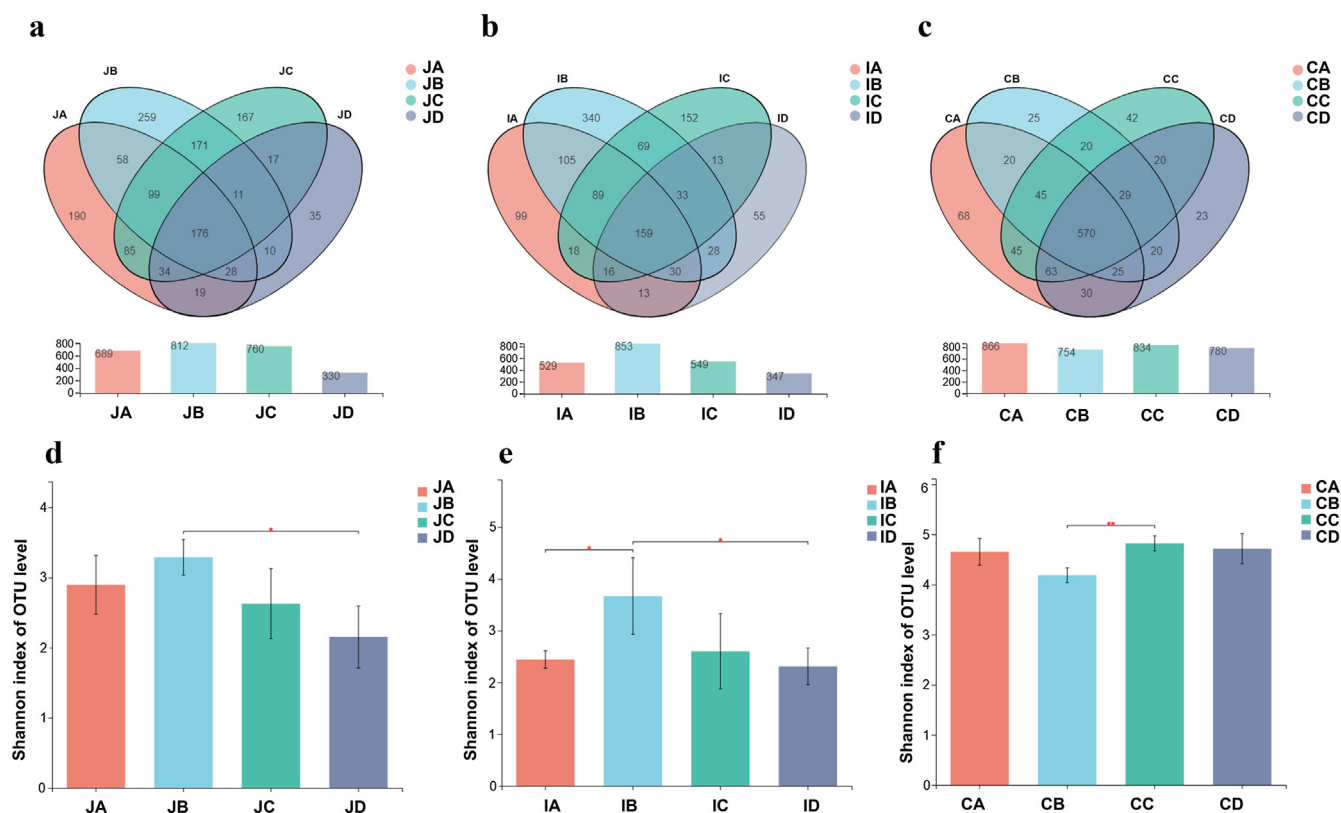


Fig. 2. The shared and specific operational taxonomic units (OTUs) in different intestinal sections when Xuefeng black-boned chicken were fed different *Macleaya cordata* extract (MCE) (a: jejunum; b: ileum; c: cecum). Shannon index of OTU level (d: jejunum; e: ileum; f: cecum). This figure shows the significant difference between samples, and marks with significant difference between the two groups ($0.01 < P \leq 0.05$ marked *, $0.001 < P \leq 0.01$ marked **). J, I, C represent different intestinal segments (J: jejunum, I: ileum, C: cecum); A, B, C, D represent different MCE addition amount groups: (A: MCE 0 mg/kg; B: MCE 100 mg/kg; C: MCE 150 mg/kg; D: MCE 200 mg/kg).

Firmicutes, *Actinobacteria*, *Proteobacteria*, *Cyanobacteria* and *Bacteroidetes* were the major microbial groups, reaching to 99.42% of jejunum. Moreover, *Firmicutes* was the dominant phylum (Fig. 4a). These results showed group C and group B were significantly higher relative abundance of *Firmicutes* compared with group A. However, a lower abundance of *Cyanobacteria*, *Proteobacteria*, and *Actinobacteria* was observed in group B, the abundance of *Firmicutes* decreased while the *Bacteroidetes*, *Cyanobacteria*, *Proteobacteria* and *Actinobacteria* increased. The community composition of group C was similar to group D, while the species distribution of group B was not similar to other groups. The main differences were higher abundance of *Actinobacteria*, *Proteobacteria*, *Cyanobacteria* and *Bacteroidetes*. The inter-group difference test showed that the top 20 phyla had no significant difference between the groups (Fig. 4d, $P > 0.05$).

As shown in Fig. 4b, the dominant bacterial phyla in the ileum were *Firmicutes*, *Proteobacteria*, *Bacteroidetes*, *Actinobacteria* and *Cyanobacteria*, including 99.40% of the microbial population (the most abundant were *Firmicutes*). Among those, compared with group A, the abundance of *Firmicutes* increased in group C and group D while in group B decreased. In MCE-treated groups, the abundance of *Proteobacteria* increased while of *Cyanobacteria* decreased. Among the four groups, the abundance of *Bacteroidetes* and *Actinobacteria* in group B was the highest. The community composition of group C was similar to group D, while the species distribution of group B was not similar to other groups. Inter-group difference test showed that there was no significant difference between the groups which conducted for the top 20 phyla in each group (Fig. 4e).

As shown in Fig. 4c, the main phyla of cecal bacteria were *Bacteroidetes*, *Firmicutes*, *Proteobacteria*, *Spirochaetae*, *Synergistetes*,

Actinobacteria and *Tenericutes*, accounting for 98.96% of the total bacteria. Furthermore, *Bacteroidetes* and *Firmicutes* were the main phyla of cecum accounting for 91.84%. Compared with group A, the abundance of *Bacteroidetes* increased with dietary MCE supplementation, while the abundance of *Firmicutes* decreased. The community composition of group C was similar to group D, while the species distribution of group B was not similar to other groups. As shown in Fig. 4f, an inter-group difference test was conducted for the top 20 phyla in each group and the results showed that the abundance of *Firmicutes* in the group A was greater than the other three groups significantly ($P < 0.01$). *Deferribacteres* in group D were more than the other three ($P < 0.05$). No significant difference was found between other phyla ($P > 0.05$).

Heat map analysis-genus level

As shown in Fig. 5a, the dominant genus of jejunum was *Lactobacillus*. And compared with group A (Fig. 5d), *Lactobacillus* in group D had a significant increase trend ($P < 0.1$). The community composition of group A was similar to group C, while the species distribution of group B were not similar to other groups (Fig. 5a). Compared with the group A (Fig. 5d), *Aeriscardovia* in group D had a significant increase trend ($P < 0.1$). No significant difference was found in other genera ($P > 0.05$). As shown in Fig. 5b, the bacteria in the ileum with the highest abundance was *Lactobacillus*. The community composition of group A was similar to group D, while the species distribution of group B were not similar to other groups. Compared with group A (Fig. 5e), *Lactobacillus* in group A had a significant increase trend ($P < 0.1$). There was no significant difference among other genera ($P > 0.05$). As shown in Fig. 5c, the dominant genus of cecum was *Bacteroides*. The community compo-

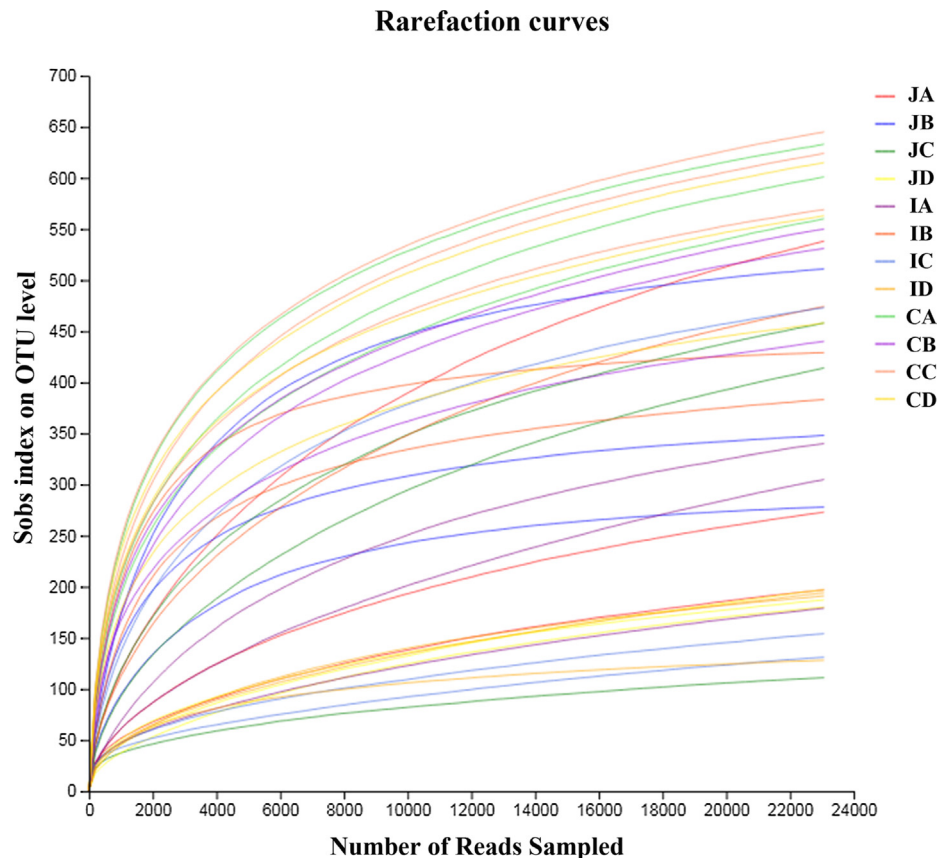


Fig. 3. Rarefaction Curve of intestinal content samples when Xuefeng black-boned chicken were fed different *Macleaya cordata* extract. The abscissa represents the amount of sequencing data randomly drawn; the ordinate, the number of species observed (Sobs index). OTUs represent operational taxonomic units; J, I, C represent different intestinal segments: (J: jejunum, I: ileum, C: cecum); A, B, C, D represent different *Macleaya cordata* extract (MCE) addition amount groups: (A: MCE 0 mg/kg; B: MCE 100 mg/kg; C: MCE 150 mg/kg; D: MCE 200 mg/kg).

sition of group B was similar to group D, while the species distribution of group A were not similar to other groups. There were no significant differences between genera (Fig. 5f, $P > 0.05$).

Principal coordinate cluster analysis

Weighted unifracs distance was calculated based on the OTU species and relative abundance of the samples. Principal coordinate analysis (PCoA) clustering diagram was drawn by using the distance matrix, and the inter-group similarity was analysed (Fig. 6). The results showed that group A was similar to group B, and group C was similar to group D in the jejunum. The samples were clustered with each other according to the different groups, indicating that the similarity within the group was higher than the similarity between the groups. In addition, the degree of clustering of individuals in group D was high, indicating that the similarity within group D was higher than other groups (Fig. 6a). In the ileum, group A was not similar compared to the other three groups, while the similarity between groups B, C and D was higher (Fig. 6b). As shown in Fig. 6c, the similarity within group C was higher than other groups in cecum.

Discussion

Natural compounds derived from *Macleaya cordata*, such as sanguinarine (one of the quaternary benzo[c] phenanthridine alkaloids), have been used as feed additive for pigs and poultry (Vieira et al., 2008; Pellikaan et al., 2010). In addition, they have gained attention as alternatives to antibiotic growth promoters

(Kim et al., 2012). The results show that dietary MCE can improve performance, intestinal health, immunity and growth of piglets (Bojjireddy et al., 2013; Khadem et al., 2014). In addition, an average daily dose of oral alkaloids of no more than 5 mg/kg animal BW is secure (Kosina et al., 2004). Numerous trials have shown that sanguinarine has a positive effect on intestinal health. Sanguinarine could inhibit noxious bacteria in intestinal wall, diminished the production of venenous compounds and harm to intestinal epithelial cells, so as to protect the intestinal mucosa (Yakhkeshi et al., 2011). Moreover, sanguinarine could improve intestinal barrier functions of growing piglets (Liu et al., 2016a). Many studies have shown that sanguinarine and chelerythrine extracted from *Macleaya cordata* could promote the growth of the intestinal villi and increase the area of intestinal villi absorption (Chen et al., 2018a; 2018b). The intestinal mucosal barrier function was enhanced by the expression of tight junction protein in the intestine (Liu et al., 2016a). In addition, MCE regulated the intestinal microbial flora, promoted the proliferation of *Lactobacillus*, and inhibited the reproduction of *Colibacillus* (Huang et al., 2018), improved the digestibility of nutrients in the diet, and promoted the absorption of nutrients (Boroojeni et al., 2018). We evaluated the effect of MCE levels on intestinal morphology and microbial diversity in different intestinal segments of Xuefeng black-boned chicken.

Good intestinal morphology was a guarantee of intestinal health (Zang et al., 2009). The intestinal mucosal morphology was the key to the digestion and absorption of nutrients, and jejunum and ileum determined the main efficiency of nutrient absorption (El Aidy et al., 2015). In particular, the small intestinal VH, CD,

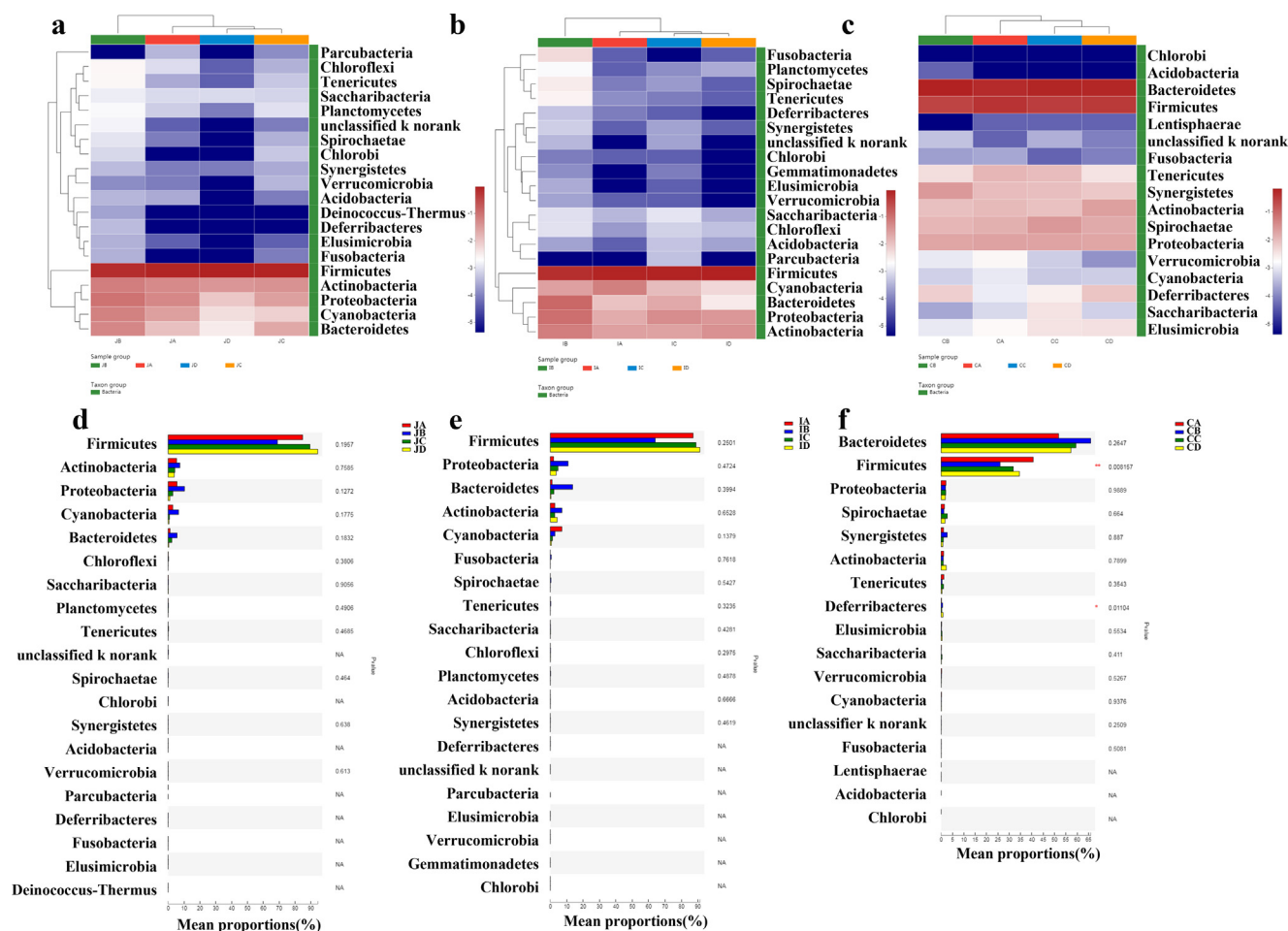


Fig. 4. The Heat map chart of the top 20 phyla in different intestinal segments when Xuefeng black-boned chicken were fed different *Macleaya cordata* extract (MCE) (a: jejunum; b: ileum; c: cecum). Comparison of differences between groups of the top 20 phyla (d: jejunum; e: ileum; f: cecum). One-way ANOVA, * $P < 0.05$, ** $P < 0.01$; A: *Macleaya cordata* extract (MCE) 0 mg/kg; B: MCE 100 mg/kg; C: MCE 150 mg/kg; D: MCE 200 mg/kg.

and VH/CD values were important indicators for measuring the small intestinal digestion and absorption function. The ratio of VH/CD comprehensively reflected the function of the small intestine, which increased, indicating that the digestion and absorption function of the small intestine was enhanced and the intestinal structure was improved (Sen et al., 2012). Our data showed that the CD of jejunum was significantly lower by the supplementation of MCE ($P < 0.05$), and the VH/CD value of 200 mg/kg group was the highest ($P > 0.05$). Karimi et al. (2014) confirmed that different levels of Sangrovit had no significant effect on the small intestine morphology of broilers. Compared with the standard diet, the supplementation of sanguinarine in the diet of hens could cause the jejunal crypt to become shallow and the VH/CD value to increase (Bavarsadi et al., 2016). The mean CD of jejunum of piglets fed the sanguinarine diet was lower (Chen et al., 2018a; 2018b), which was similar to our experimental results. It was suggested that adding 200 mg/kg of MCE could improve the absorption capacity of jejunum.

Gut was the most important and complex micro-ecosystem in organisms. Intestinal microorganisms and their metabolites were involved in lots of physiological activities, such as nutrient metabolism and immune regulation (Stanley et al., 2014). The distribution of intestinal microflora in duodenum, jejunum, ileum, cecum and rectum was not identical (Huang et al., 2018). The community structure of the intestinal microflora was influenced by many factors, such as heredity, diets, and geography (Contreras et al., 2012;

Godoy-Vitorino et al., 2012). In our study, 16S rDNA gene sequencing was employed to detect the microflora of jejunum, ileum and cecum of Xuefeng black-boned chicken supplied with dissimilar doses of MCE diets. The sharing and specific OTUs of different intestinal segments in different groups were showed in the Venn diagram. Dosage of 100 mg/kg MCE in the diet had the most unique OTUs (259 and 340 in jejunum and ileum). And the OTU number of dietary 100 mg/kg MCE supplementation in jejunum was higher than dietary 200 mg/kg group, at the same time, the number of OTU of dietary 100 mg/kg MCE in ileum was higher than the other three groups. These implied that the addition of 100 mg/kg MCE to the diet increased the microbial diversity of jejunum and ileum. Possibly due to the antibacterial effect of sanguinarine, the microbial diversity with the dietary 200 mg/kg MCE decreased. Moreover, it has reported that the diversity and function of intestinal microflora in chickens were almost related to the microflora in ileum and cecum (Bjerrum et al., 2006). Shannon index showed that the diversity of cecal microflora was higher than jejunum and ileum, which was similar to the results of former studies (the diversity of cecal microflora was higher than ileal microflora) (Dumonceaux et al., 2006; Mohd Shaufi et al., 2015).

Firmicutes was the dominant phylum in jejunum and ileum of broiler chicken accounting for >60% of sequences, while *Bacteroidetes* was the dominant phylum in cecum (>50% of sequences) (Xiao et al., 2016). However, in the cecum, *Bacteroidetes* was the most dominant phylum, which was followed by *Firmicutes*, which was

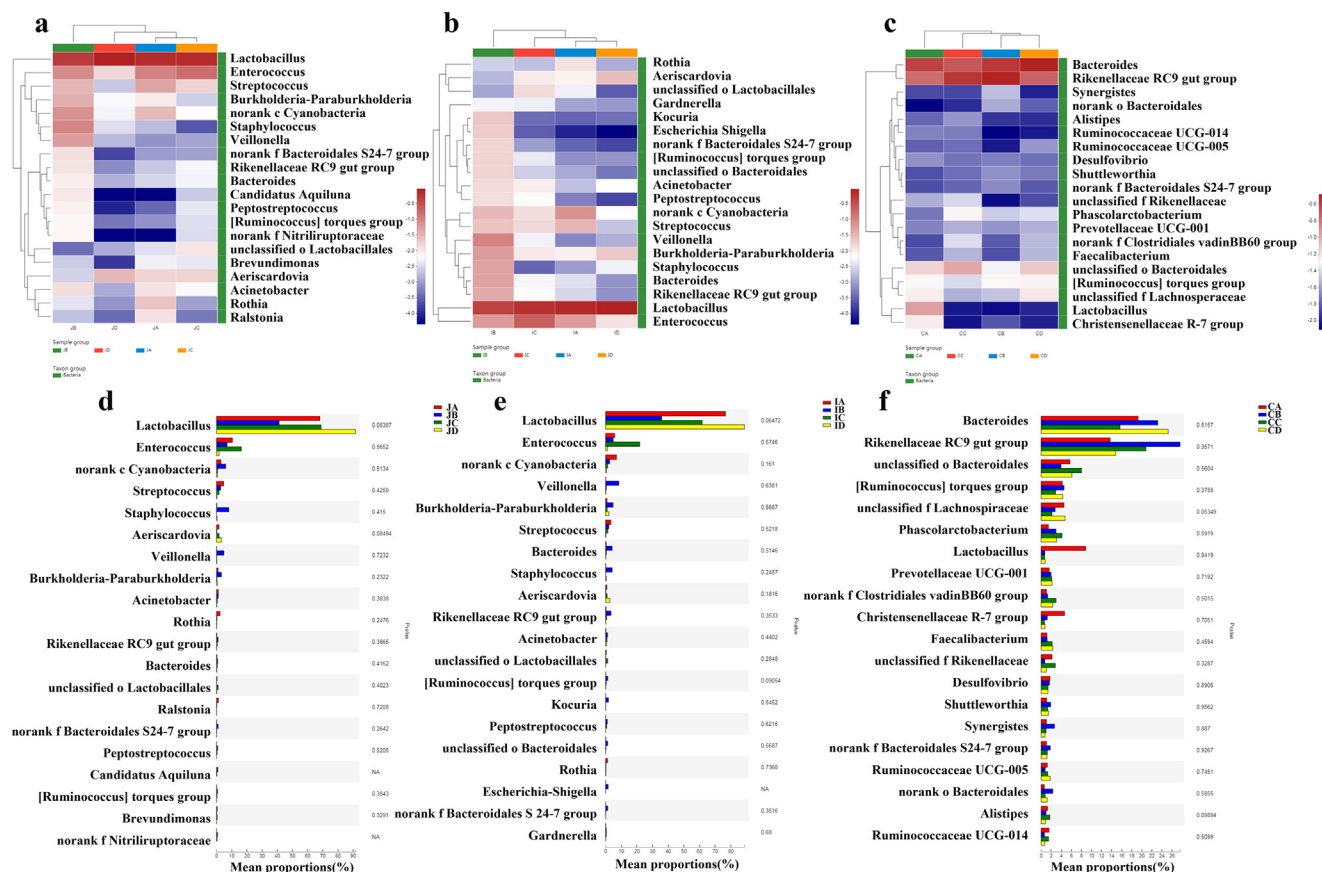


Fig. 5. The Heat map chart of the top 20 genera in different intestinal segments when Xuefeng black-boned chicken were fed different *Macleaya cordata* extract (MCE) (a: jejunum; b: ileum; c: cecum). Comparison of differences between groups of the top 20 genera (d: jejunum; e: ileum; f: cecum). One-way ANOVA; A: *Macleaya cordata* extract (MCE) 0 mg/kg; B: MCE 100 mg/kg; C: MCE 150 mg/kg; D: MCE 200 mg/kg.

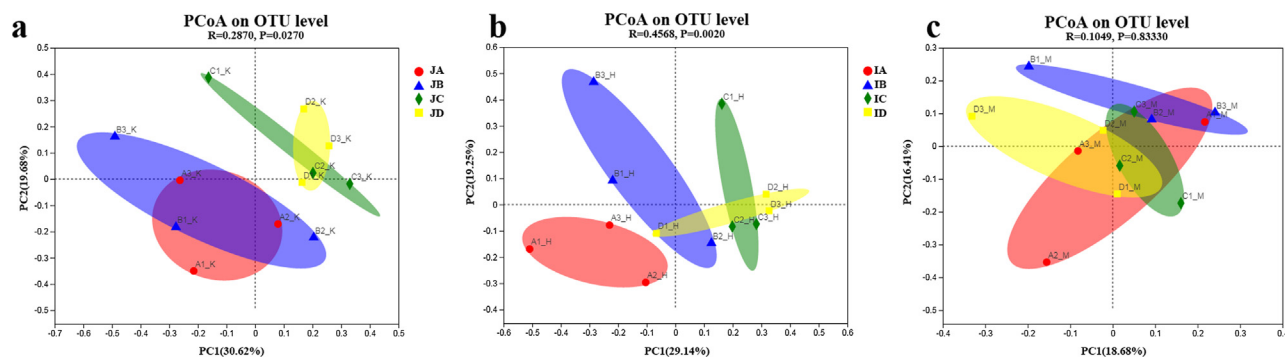


Fig. 6. Cluster analysis by principal coordinate analysis in different intestinal segments when Xuefeng black-boned chicken were fed different *Macleaya cordata* extract (MCE). a: jejunum; b: ileum; c: cecum. J, I, C represent different intestinal segments: (J: jejunum, I: ileum, C: cecum); A, B, C, D represent different *Macleaya cordata* extract (MCE) addition amount groups: (A: MCE 0 mg/kg; B: MCE 100 mg/kg; C: MCE 150 mg/kg; D: MCE 200 mg/kg). The X-axis and Y-axis represent the two selected principal axes, and the percentage represents the value of the principal axis to explain the difference in sample composition. The scale of the X-axis and the Y-axis is relative distance, which makes no sense; Dots of different colours or shapes represent samples from different populations, and the closer the two sample points are, the more similar the species composition of the two samples is.

agreeable to the result of Xiao et al. (2016). *Bacteroidetes*, which was a primarily Gram-negative bacteria, helped to digest starch within the rumen (Stevenson and Weimer, 2007). *Bacteroides* could degrade fibre and starch, which explained why *Bacteroidetes* were the dominant bacteria in the cecum. Peng et al. (2015) reported that long-term intake of fat could expand the proportion of *Firmicutes* to *Bacteroidetes*, while ingestion of dietary fibre might conduce to a decrease of the ratio. We found that, at the phylum

level, *Firmicutes* were the most abounding microorganisms in jejunum and ileum of Xuefeng black-boned chicken, followed by *Bacteroidetes*. Moreover, the abundance of *Bacteroides* was higher and *Firmicutes* was less in MCE-added group than control in the cecum. *Deferribacteres* were iron-reducing bacteria, isolated from water samples, which participated in detoxification or bioremediation of minerals in the environment so that they were not used as food sources by their hosts (Magali and Marie-Anne, 2003). *Defer-*

ribacteres was almost only presented in cecum (Xiao et al., 2016). It was found that, in cecum, *Deferribacteres* increased with dietary 200 mg/kg MCE supplementation in our study.

At the genus level, the results showed that the main bacteria in three intestinal segments were *Lactobacillus*, *Enterococcus*, *Bacteroides* and *Streptococcus*. *Lactobacillus* and *Enterococcus* were the most common in jejunum and ileum and less in cecum, which was coherent with the results of Bjerrum et al. (2006) and Lu et al. (2003). *Lactobacillus* has been reported to be involved in the production of some essential vitamins and organic acids (Neal-McKinney et al., 2012; LeBlanc et al., 2013), indicating that *Lactobacillus* had a significant contribution to nutritional absorption and related intestinal function. *Lactobacillus* was the predominant genus which was influenced by MCE. Our study showed that the content of *Lactobacillus* of the 200 mg/kg MCE group was greater than others in jejunum and ileum, suggesting that the addition of 200 mg/kg MCE would help the absorption of nutrients. Moreover, increased *Lactobacillus* probably caused competitive exclusion against harmful bacteria but provided a free place for colonisation of beneficial bacteria, which decreased directly or indirectly the microbial diversity.

Aeriscardovia belongs to the *Bifidobacterium* family which was one of the most important physiological bacteria in intestinal tract of human and animals. It could produce short-chain fatty acids, vitamins, bacteriocins and antibiotics. Also, it has numerous important physiological functions, such as maintaining microecological balance, biological antagonism, immune regulation, nutrition, synergistic degradation of polysaccharides (Sonnenburg et al., 2006). In the present study, with 200 mg/kg MCE addition, *Aeriscardovia* presented an increasing trend compared with the control in jejunum. However, no significant increase of *Aeriscardovia* was observed in ileum.

In the foregut, MCE inhibited genera such as *Streptococcus*, *Rothia* and pathogens which might pose a serious threat to animal health. However, the regulatory effect of MCE might not relate to common pathogens such as *Salmonella* or *Campylobacter*, which was not observed in our results. As in previous studies (Stanley et al., 2014), *Lactobacillus* was the dominant genus of the foregut. And in our results, *Bacteroides* were dominant in cecum, as same as *Ruminococcus* in cecum. This was accordant with results of Xiao et al. (2016) and Dumonceaux et al. (2006). The addition of MCE to diet did not change dominant flora of each intestine of Xuefeng black-boned chicken.

The results of PCoA analysis showed that microbial community between control group and addition of MCE group was different in jejunum and ileum, which indicated that dietary addition of MCE changes intestinal microbial composition. However, the similarity of microbial community between the four groups was higher in the cecum. Our results showed that MCE had less effect on microbial community composition of the cecum.

Conclusion

In this study, we found that dietary MCE supplementation had beneficial effects on improved intestinal morphology and gut microbiota composition of Xuefeng black-boned chicken from day 329 to day 420 evidenced by increased beneficial bacteria,

increased bacterial community diversity and relative abundance in jejunum and ileum. These findings provided new insights into suitable alternatives to antibiotics in poultry production.

Ethics approval

The study was authorised by the Animal Care and Use Committee of Hunan Agricultural University (HUNAU2020003).

Data and model availability statement

The datasets of this study can be found in online repositories. The names of the repository/repositories and accession number (s) can be discovered below: <https://www.ncbi.nlm.nih.gov/bio-project/PRJNA766322/>.

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Author contribution

SCG and XYZ designed the experiment. LLL and CQH performed the experiment and XYQ, PL and FLL collected samples. BX, SWZ and QQG analyzed samples and analyzed the data. JXL, SGT and CQH conceptualized the paper, compiled all of the information and prepared the manuscript. SCG, JXL and XYZ conceptualized the paper, provided insights to the entire manuscript and contributed to the writing. All authors read and approved the final manuscript.

Declaration of interest

None of the authors has any conflict of interest to declare.

Acknowledgement

None.

Financial support statement

This study was funded by the start-up funds from Hunan Agricultural University and National Natural Science Foundation of China (32 072 711).

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